Qingzheng Yu

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EDUCATION

Ph.D. in Astrophysics *Sep. 2017 - Jun. 2023*
Department of Astronomy, Xiamen University, Fujian, China
– Thesis: *On the Cold Gas Evolution of Interacting Galaxies*
– Advisor: Professor Taotao Fang

Bachelor of Science in Physics *Sep. 2012 - Jun. 2016*
Department of Physics, Guangzhou University, Guangdong, China

PROFESSIONAL APPOINTMENTS

Postdoctoral Research Fellow *Sep. 2024 - present*
Department of Physics and Astronomy, University of Florence, Italy

Postdoctoral Research Assistant *Jul. 2023 - Sep. 2024*
Department of Astronomy, Xiamen University, Fujian, China

Research Assistant *Sep. 2017 - Jun. 2023*
Department of Astronomy, Xiamen University, Fujian, China

RESEARCH INTERESTS

Galaxy Formation and Evolution • Circumgalactic Medium • Interstellar Medium,
Gas Recycling and Star Formation of Galaxies • Galaxy Interactions and Mergers,
Feeding and Feedback of AGN • H I Absorption Survey toward Radio AGNs

ACCEPTED PROPOSALS

PI, ∼91 h, ALMA ACA 2026
Projects: 2026.1.01015.S, 2026.1.01414.S

PI, ∼263.5 h, Green Bank Telescope (GBT) 2026B
Projects: GBT26B-120

PI, ∼60 h, ALMA ACA 2025
Projects: 2025.1.01510.S

PI, ∼206 h, Five-hundred-meter Aperture Spherical radio Telescope (FAST) 2020-2026
PID: PT2020_0152, PT2021_0067, PT2022_0090, PT2023_0004, PT2023_0021, PT2024_0216, PT2024_0244, PT2025_0202, PT2026_0204

PI, 19.7 h, Karl G. Jansky Very Large Array (JVLA) 2023A
Projects: 23A-236

PI, 57 h, Institut de Radioastronomie Milimetrique (IRAM) 30 m 2021-2022A
Projects: E01-21, 029-22

PI, 21 h, James Clerk Maxwell Telescope (JCMT) 2021B
Projects: M21BP051

PI, 4 nights, P200/Hale Telescope: CWI, WaSP TAP 2021A-2024A
Projects: CTAP2021-A0034, CTAP2021-B0019, CTAP2024-A0020

PI, 11 h, Green Bank Telescope (GBT) 2021A
Projects: GBT-21A-245

Co-I, 41.4 h, Karl G. Jansky Very Large Array (JVLA) 2024A
Projects: 24A-407

Co-I, ∼134 h, Five-hundred-meter Aperture Spherical radio Telescope (FAST) 2019-2022
2019a-005-S, PT2021_0147, PT2021_0186, PT2021_0120, PT2021_0139, PT2021_0040, PT2022_0181, PT2024_0044, PT2024_0106

OBSERVING EXPERIENCE

Radio single-dish spectroscopy and mapping:

FAST, H I spectral line observations of nearby galaxies, pool, 94 h
FAST, H I mapping of nearby galaxy groups, pool, 34 h
FAST, H I mapping of high-velocity clouds, pool, 64 h
GBT, H I mapping of high-velocity clouds, remote, 20h
GBT, H I spectral line observations of nearby galaxies, pool, 11h
IRAM 30m, CO spectral line observations of nearby galaxies, remote, 149 h
JCMT, CO spectral line observations of nearby galaxies, pool, 21 h
Parkes, H I spectral line observations of nearby galaxies, remote, 22 h

Radio interferometer:

JVLA, H I spectral line and continuum imaging of nearby galaxies, pool, 19.7 h

Optical imaging and spectroscopy:

P200/Hale, narrow-band imaging of nearby galaxies with WaSP, remote, 2 nights
P200/Hale, spectroscopic observations of nearby galaxies with DBSP, remote, 1 night

SELECTED CONFERENCE CONTRIBUTION

Mpc-scale H I Flows in the Neighborhood of HCG 100 07/2026
-Contributed talk on “Many Scales in Galaxy Enviroments”, Ascona, Switzerland

Hidden H I Clouds in the MW’s Nuclear Wind 05/2026
-Contributed talk on “Traversing the Galactic Centre in Space and Time”, Brno, Czech Republic

Tracing Cold Gas Evolution in Major-merger pairs and Compact Groups 06/2025
-Contributed talk on “Dancing in the Dark: When Galaxies Shape Galaxies”, Sexten, Italy

Hidden H I Clouds in the MW’s Nuclear Wind 05/2025
-Contributed talk on “Contents of the Fermi Bubbles and Related Topics”, Green Bank, USA

Exploring the Baryon Cycle through Interacting Galaxies and AGNs 05/2024
-Invited talk for seminar at Beijing Planetarium, Beijing, China

Cold Gas Evolution of Major-merger Pairs in MaNGA survey 08/2024
-Contributed talk on "Seventh Annual SKA Science Symposium", Shanghai, China

H I Absorption in Low-power Radio AGNs Detected by FAST 10/2023
-Contributed talk on "Galaxy Formation in Hangzhou: Observations and Physics of AGN Feedback", Hangzhou, China

H I Observations of Low-power Radio AGNs and the Magellanic Stream with FAST 09/2023
-Contributed talk on "The First FAST Science Symposium", FAST site, Guizhou, China

H I Absorption in Low-power Radio AGNs Detected by FAST 07/2023
-Contributed talk on "Symposium on Multiwavelength Studies of Quasars and AGNs", Lijiang, China

Investigating Galaxy Interactions and AGN through H I Observations 06/2023
-Contributed talk on "Sixth Annual SKA Science Symposium", Shanghai, China

On the H I Content of MaNGA Major Merger Pairs, 2023, IAU Symposium, 373, 186 08/2022
-Contributed remote talk on "IAUGA 2022 Symposium 373", Busan, Korea

On the H I Content of MaNGA Major Merger Pairs 12/2021
-Contributed talk on "Jing-Guang-Xia Astrophysics Symposium", Xiamen, China

H I observations of MaNGA merging galaxies and HVCs with FAST 04/2021
-Contributed talk on "CRAFTS and FAST data analysis workshop", Nanjing, China

TEACHING AND MENTORING

Teaching at Xiamen University

Observational Astronomy (Graduate course), Lecturer (remote), ∼30 students 2025 Spring
Observational Astronomy (Graduate course), Lecturer (1 week), ∼30 students 2024 Spring
Observational Astronomy (Graduate course), Lecturer (1 week), ∼30 students 2023 Spring
• Introduced the H I and extragalactic observations in radio band
Observational Astrophysics (undergraduate course), Teaching Assistant, ∼50 students 2018 Spring
• Built course websites for problem sets, solutions, and FAQ's for sessions

Summer school and workshop

The 11th Square Kilometre Array (SKA) Summer School, Lecturer (remote), ∼50 students 2026
• Introduced the deep FAST observations and analysis of diffuse H I

Student Research Project Supervising

Zeqian Xu (XMU '25→ Master@Leiden U): H I Observations with FAST, co-advisor 2022 - 2025
Yang Su (XMU '26 Master): A FAST Survey of H I Absorption Lines, co-advisor 2023 - 2026
• one ApJ paper published in 2026
Shulan Yan (XMU '26 PhD): Observations of nearby galaxies with FAST, co-advisor 2024 - 2026
• one ApJ paper published in 2026
Yuxin Xiao (XMU '25→ PhD@XMU): HVC Observations with FAST, co-advisor 2022 - present
• one ApJ paper published in 2026

COMMUNITY SERVICE

Review panel member, FAST medium-size proposal, 2025 -2026

Committee Member, FAST User Committee, ∼20 out of ∼500 users 2021 - present

Organizer, student discussion with colloquium speaker, Xiamen University 2021 - 2024

Organizer, group meeting for the Galaxy and Cosmology Group, Xiamen University 2020 - 2024

OUTREACH

Technical support, Notti d'Estate, Firenze, Italy 2025 -2026

SKILLS

Languages: Chinese (Mother Language), English (Professional), Italian (A2)
Programming: Python, IDL, Bourne Shell, Matlab
Software & Tools: GILDAS, HIFAST, Starlink, MIRIAD, IRAF, CWITools, GBTIDL, CASA

AWARDS/FELLOWSHIPS

2024-2027: European Research Council (ERC) postdoctoral fellowship @ University of Florence
2024-2027: Kavli Institute of Astronomy and Astrophysics (KIAA) fellow (declined)
2013: Undergraduate scholarship from NAOC

REFERENCES

• Taotao Fang
Email: fangt@xmu.edu.cn
Professor of Astronomy & Astrophysics, Department of Astronomy, Xiamen University

• Cong Kevin Xu
Email: coxu@ipac.caltech.edu
Professor of Astronomy & Astrophysics,
Chinese Academy of Sciences South America Center for Astronomy,
National Astronomical Observatories, Chinese Academy of Sciences

• Enrico Di Teodoro
Email: enrico.diteodoro@unifi.it
Professor of Astronomy & Astrophysics, Department of Physics and Astronomy, University of Florence

• Junfeng Wang
Email: jfwang@xmu.edu.cn
Professor of Astronomy & Astrophysics, Department of Astronomy, Xiamen University

PUBLICATIONS

19 published peer-reviewed papers listed below; seven are first- or corresponding-authored. A dagger (†) denotes corresponding author.

• 1st-/corresponding-authored (†):

7. Xiao Y.,**Yu Q.**†, Fang T., Li D., et al., *FAST Observations of Filamentary and Compact HI Structure in a Magellanic Stream IV Field*, 2026, ApJ, 1011, 27
6. **Yu Q.**, Fang T., Di Teodoro E. M., Cheng C., Xu C. K., *Megaparsec-Scale Neutral Hydrogen Flows in the Neighborhood of Hickson Compact Group 100*, 2026, ApJL, 1007, L32
5. Su Y., **Yu Q.**†, Fang T., Wang J., Wu J., Zhang B., *A FAST Survey of HI Absorption in Low-power Radio Sources*, 2026, ApJ, 1000, 108
4. Yan S., Ma A., **Yu Q.**†, Fang T., He C., Zhu M., *HI Content of Group Galaxies from the FAST All Sky HI Survey* , 2026, ApJ, 997, 254
3. **Yu Q.**, Xu C. K., Feng S., Feng S., Gao Y., Jiang X.-J., et al., *CO Observations of Early-mid Stage Major-mergers in MaNGA Survey*, 2024, ApJS, 273, 2
2. **Yu Q.**, Fang T., Wang J., Wu J., *HI Absorption in Low-power Radio AGNs Detected by FAST*, 2023, ApJ, 952, 144
1. **Yu Q.**, Fang T., Feng S., Zhang B., Xu C. K., Wang Y., Hao L., *On the HI Content of MaNGA Major Merger Pairs*, 2022, ApJ, 934, 114

• Co-authored:

12. Yan S., Fang T., Jiang X., Wang J., **Yu Q.**, et al., *Redshifted Cool Gas in the Disturbed HI Disk of the Starburst Dwarf NGC 4214*, 2026, ApJL, 1011, L43
11. Xu Y., Huang F., Sun M., Fang T., Wu J., **Yu Q.**, et al., *Radio-excess Active Galactic Nuclei in Dwarf Galaxies: Multiwavelength Diagnostics from Optical Spectroscopy to Mid-infrared and Variability Analysis*, 2026, ApJ, 1005, 181
10. Zhang Z., Zhang X., Fang T., +7+ **Yu Q.**, et al., *Probing the Hot Gaseous Halos of Milky Way-like Galaxies in the TNG50 Simulation*, 2026, ApJ, 1004, 184
9. Yan S., **Yu Q.**, Fang T., He C., Ma A., Zhu M., et al., *HI Gas and Star Formation in Major Galaxy Pairs from the FAST All Sky HI Survey (FASHI)*, 2026, ApJ, 1002, 183
8. Di Teodoro E. M., Heyer M., Krumholz M. R., Armillotta L., +7+ **Yu Q.**, *A survey of molecular clouds in the Galactic center's outflow*, 2026, A&A, 706, A330
7. Zhang Z., Zhang X., Li H., Fang T., +5+ **Yu Q.**, et al., *Tracing the Origins of Hot Halo Gas in Milky Way-type Galaxies with SMUGGLE*, 2025, ApJ, 991, 170
6. Hao L., Li Z., Shen F., Xu Y., Huang Y., Lee K.,**Yu Q.**, et al., *A Linear Decomposition Method to Analyze and Study Pulsar Mode Changes*, 2025, ApJ, 983, 63
5. Cao T.-W., Li Z.-J., Chen P.-B., Zhang C.-Y., Galaz G., Cheng C., **Yu Q.**, et al., *Ionized and Cold Gas Components in Low Surface Brightness Galaxy AGC 102004*, 2024, ApJ, 971, 181

4. He C., Xu, C., Lisenfeld, U., Dai, Y.∼S., Fang T., Huang J., Wang W., **Yu Q.**, *Close Major-merger Pairs at z = 0: Star-forming Galaxies with Pseudobulges*, 2024, RAA, 24, 055005

3. Zhang Z., Zhang X., Li H., Fang T., **Yu Q.**, et al., *Low- and High-velocity O VI in Milky Way-like Galaxies: the Role of Stellar Feedback*, 2024, ApJ, 962, 15

2. Yi T., +17+ **Yu Q.** +10, *A Dynamically discovered and characterized non-accreting neutron star-M dwarf binary candidate*, 2022, Nature Astronomy, 6, 1203

1. Zhang B., Zhu M., Wu Z.-Z., **Yu Q.-Z.**, et al., *Extragalactic HI 21-cm absorption line observations with the Five-hundred-meter Aperture Spherical radio Telescope*, 2021, MNRAS, 503, 5385